

Extra Problems

Please work on the problems below and then discuss the problems at the [Week 2 Extra Problems Discussion](#) [↗](#) forum.

Extra problems week 2:

In what follows $n.G$ or nG represents the sum of n copies of G , e.g., $3G = G + G + G$, similarly $-n.G$ or $-nG$ is the sum of n copies of $-G$.

Games:
Hackenbush
A One blue stalk B. One blue stalk with 1 red on top C. One blue with 3 red on top

Nim
D. nim heap of size 1 E. Nim heap of size 3

Cutcake:
F. 3 rows, 4 columns G 3 rows 6 columns

Run Over
H LR
I. __LR
J. LLR
K LRRR

Who wins? L first? R first?

- 1. $A - 6C$
- 2. $A - 9C$
- 3. $C - 5B$
- 4. $H + A$
- 5. $H + 2A$
- 6. $H + 2A - B$
- 7. $H + 2A - B + E$
- 8. $F - A$
- 9. $F - 2A$
- 10. $F - 3A$
- 11. $G - A$
- 12. $G - 2A$
- 13. $G - 4B + D$
- 14. $H + A$
- 15. $H + 2H$
- 16. $H + A + B$
- 17. $J - H + 2C$
- 18. $4I - 9A$
- 19. $4I - 11A$
- 20. $K + 8A$

MAKE UP YOUR OWN!

